Python Programming with GenAI

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Agenda

- Lists and Tuples
- Looping Statements
- Functions

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Let's begin the discussion by answering a few questions on the fundamentals of Python programming

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Python Programming with GenAI Quiz

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POWER AHEAD

Consider a customer database for managing Netflix subscriptions. Each customer has a unique ID, name, email, and subscription type, which remain constant once registered.

What data structure would be the best choice to store this information?

A. Only List

B. Only Tuple

C. List and Tuple

D. List and Dictionary

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Data Structures

List	Tuple	Dictionaries
A collection of items of any data type	A collection of items of any data type	A collection of key-value pairs
Mutable (can be modified)	Immutable (cannot be modified)	Mutable (can be modified)
Syntax: `mylist = ["Element 1", "Element 2", "Element 3"]`	Syntax: `mytuple = ("Element 1", "Element 2", "Element 3")`	Syntax: `mydict = {1: 'Element 1', 2: 'Element 2', 3: 'Elements 3'}`
Example: `X=["a", 2, True, "b"]`	Example: `X=("a", 2, True, "b")`	Example: `X={1:'Jan', 2:'Feb', 3:'Mar'}`

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Which of the following will retrieve the two middle elements from the list?

```
my_list = [1, "two", 3.5, "four", 5, 6.0, "seven", 8.2]
```

A `my_list[3:5]`

B `my_list[4:6]`

C `my_list[-5:-3]`

D `my_list[-4:-6]`

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Lists - Slicing

To retrieve a specific **subset of elements from a list**, one can **slice the list by index**

The format for list slicing is as follows:

```
<list_name>[Start Index : End Index : Index-Jump]
```

Consider a list `my_list = ['a', 'b', 'c', 'd']`

Negative Index	-4	-3	-2	-1
Elements	a	b	c	d
Index	0	1	2	3

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Which of the following statements is true regarding loops in Python?

A. The "for" loop requires a condition to be evaluated before execution

B. The "for" loop iterates through a sequence, executing on each element

C. The "while" loop requires a condition to be evaluated before execution

D. The "while" loop iterates through a sequence, executing on each element

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Looping Statements

Used to **repeat a single statement or a set of statements**

Looping statement	Syntax	Example	Output
for	`for iter var in seq:` `  statements(s)`	`for i in range(1, 5):` `  print(i)`	Prints all integers from 1 till 5 (excluded)

while	`while condition:`
	`  statement(s)`
		`i = 1`
		`while i < 5:`
		`  print(i)`
		`  i += 1`
			Prints all integers from 1 till 5 (excluded)

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Python Programming with GenAI Quiz

What is the purpose of functions in Python?

A. To execute a specific task only once in a program

B. To combine multiple variables into a single variable

C. To break code into modular chunks for reusability and organization

D. To declare built-in variables that are predefined in Python

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Functions

Block of instructions that performs a specific task

Break code into **modular chunks** which can be **reused later**

Makes the code more organized and manageable

There are two types of functions in Python

Built-in Functions: Pre-defined in Python (`print()`, `len()`, `sum()`, etc)

User-defined Functions: Defined by users to perform a specific task

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Python Programming with GenAI Quiz

The total amount payable at a supermarket after applying a relevant discount based on a customer's total purchase amount is defined by the following function:

```
def calculate_amount_payable(total_purchase):
    if total_purchase >= 5000:
        discount = 0.20
    elif total_purchase >= 3000:
        discount = 0.15
    else:
        discount = 0.10
    return total_purchase*(1 - discount)
```

Which of the following is the correct way to call this function and store the final price in a variable?

A
```
final_price = calculate_amount_payable(4000)
```

B
```
final_price = calculate_amount_payable()
```

C
```
final_price = calculate_amount_payable(4000, 15)
```

D
```
calculate_amount_payable = final_price(4000)
```

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final_price =
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```

D
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calculate_amount_payable =
final_price(4000)
```

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User-defined Functions

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Syntax for a user defined function:

```
def function_name(parameters):
    statement(s)
    return statement
```

The **`return` statement is optional** and can be used when one or more values have to be returned by the function

A function can have **multiple return statements**

One can **store the output of a function** by **assigning it to a variable**

```
variable_name = function_name(parameters)
```

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Happy Learning !



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